University Roll No._____

B. Tech. 2nd Yr. (Even Sem.), End Term Examination, 2018-19

Subject: English for Professional Purposes-II
Subject Code: BELH0004
Time: 80 Minutes Max. Marks: 20

ORAL SKILLS

Notes:

- **Answer any 4 questions out of 5 questions from Section A and Section B each.**
- **The word limit of answers must be 60 words (approx) for Section A and 120 words (approx) for Section B.**

Section A 2×4=8

1. Make the first four slides (excluding the title slide) on the topic, *Global Warming*.
2. You are a senior correspondent of The Times of India. You have taken an interview of the Director of ISRO. Write six questions of the interview.
3. Write a note on any one of the following:
   i. Effective and Ineffective Listening
   ii. Differences between Listening and Hearing
4. Write a note on Voice Modulation.
5. Read the sentences given below and identify the tone of the speaker.
   i. Offence is the best defence.
   ii. What can I do for you?
   iii. Is he going to school?
   iv. Open the door.

Section B 3×4=12

1. You are Shivam Joshi, Deputy Librarian of GLA University, Mathura. Write an order letter to the Sales Manager, S. Chand & Co., Daryaganj, New Delhi – 110002 placing an order for the books that you need.
2. You are a graduate and are applying for the position of Assistant Engineer. Write your career objective, educational qualifications and technical skills in the resume.

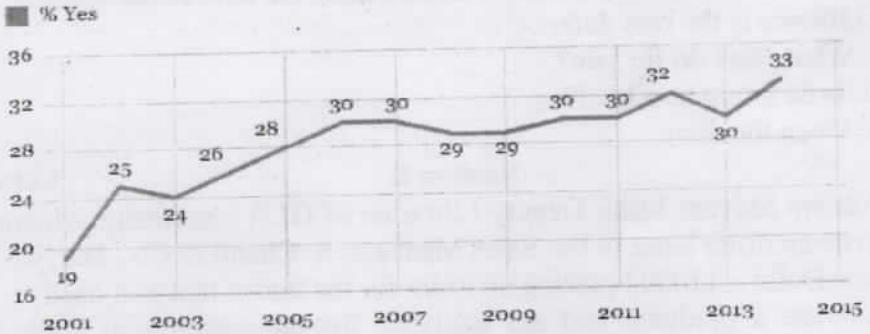
3. Write a précis of the given paragraph:

The man who is perpetually hesitating which of the two things he will do first, will do neither. The man who resolves, but suffers his resolution to be changed by the first counter-suggestions of a friend, - who fluctuates from opinion to opinion, from plan to plan, and veers like a weather-cock to every point of the compass, with every breath of caprice that blows - can never accomplish anything great or useful. Instead of being progressive in anything, he will be at best stationary, and more probably retrograde in all. It is only the man who first consults wisely, then resolves firmly, and then executes his purpose with flexible perseverance, undismayed by those petty difficulties which daunt a weaker spirit that can advance to eminence in any line. Take your course wisely, but firmly; and having taken it, hold upon it with heroic resolution, and the Alps and Pyrenees will sink before you. (154 words)

4. Give reasons to support the argument: "Good listening power is essential for a good speaker".
5. The graph below shows the percentage of Americans putting off medical treatment because of cost from 2001 to 2014. Read the graph carefully and describe it by using appropriate vocabulary.

Percentage of Americans Putting Off Medical Treatment Because of Cost

Within the last 12 months, have you or a member of your family put off any sort of medical treatment because of the cost you would have to pay?



GALLUP

SET-A

University Roll No._____

End Term Examination, 2018-19

B. Tech. 2nd Year (Even Semester)

Subject: English for Professional Purposes II

Subject Code: BELH0004

Time: 100 Minutes

Max. Marks: 30

VERBAL APTITUDE

(Answer all the questions)

Read the passages given below and answer the questions that follow.

Passage 1

If a supernova (the explosion of a massive star) triggered star formation from dense clouds of gas and dust, and if the most massive star to be formed from the cloud evolved into a supernova and triggered a new round of star formation, and so on, then a chain of star-forming regions would result. If many such chains were created in a differentially rotating galaxy, the distribution of stars would resemble the observed distribution in a spiral galaxy. This line of reasoning underlies an exciting new theory of spiral-galaxy structure. A computer simulation based on this theory has reproduced the appearance of many spiral galaxies without assuming an underlying density wave, the hallmark of the most widely accepted theory of the large-scale structure of spiral galaxies. That theory maintains that a density wave of spiral form sweeps through the central plane of a galaxy, compressing clouds of gas and dust, which collapse into stars that form a spiral pattern.

1. The primary purpose of the passage is to

A. describe what results when a supernova triggers the creation of chains of star-forming regions

B. propose a modification in the most widely accepted theory of spiral-galaxy structure

C. compare and contrast the roles of clouds of gas and dust in two theories of spiral-galaxy structure

D. describe a new theory of spiral-galaxy structure and contrast it with the most widely accepted theory

E. describe a new theory of spiral-galaxy structure and discuss a reason why it is inferior to the most widely accepted theory

2. The passage implies that, according to the new theory of spiral-galaxy structure, a spiral galaxy can be created by supernovas when the supernovas are
A. producing an underlying density wave
B. affected by a density wave of spiral form
C. distributed in a spiral pattern
D. located in the central plane of a galaxy
E. located in a differentially rotating galaxy

3. Which of the following, if true, would most discredit the new theory as described in the passage?
A. The exact mechanism by which a star becomes a supernova is not yet completely known and may even differ for different stars.
B. Chains of star-forming regions like those postulated in the new theory have been observed in the vicinity of dense clouds of gas and dust.
C. The most massive stars formed from supernova explosions are unlikely to evolve into supernovas.
D. Computer simulations of supernovas provide a poor picture of what occurs just before a supernova explosion.
E. A density wave cannot compress clouds of gas and dust to a density high enough to create a star.

4. The author's attitude toward the new theory of spiral-galaxy structure can best be described as
A. euphoric
B. enthusiastic
C. concerned
D. critical
E. disputatious

Passage 2

Although European decisions during the 16th and 17th centuries to explore, trade with, and colonize large portions of the world brought tremendous economic wealth and vast geographic influence, the enormous success of European maritime ventures during the age of exploration also engendered a litany of unintended consequences for most of the nations with which Europe interacted. Due to their incredible military force, religious zeal, and uncompromising goal of profit, Europeans often imposed their traditions, values, and customs on the people with whom they traded. They frequently acted without regard to the long-term welfare

of others as their principal concern was short-term economic gain. Since many nations that traded with Europe placed high value on their historical customs, some natives became deeply disconcerted by the changes that occurred as a result of European power. These factors, coupled with perennial domestic political instability, caused numerous countries to grow increasingly resistant to European influence. One potent example of this ideological shift can be seen in the actions of the Tokugawa government of Japan. In its Seclusion Edict of 1636, the government attempted to extricate cultural interactions with Europe from the intimate fabric of Japanese society. The Edict attempted to accomplish this by focusing on three areas. First, it sought to curb cultural exchange by eliminating people bringing European ideas into Japan. The Edict stated, "Japanese ships shall by no means be sent abroad....All Japanese residing abroad shall be put to death when they return home." Second, the Edict focused on limiting trade. Articles 11 through 17 of the Edict imposed stringent regulations on trade and commerce. Third, the government banned Christianity, which it saw as an import from Europe that challenged the long-established and well-enshrined religious traditions of Japan. The government went to considerable lengths to protect its culture. Article eight of the Edict stated, "Even ships shall not be left untouched in the matter of exterminating Christians." With the example of Japan and the examples of other countries that chose a different response to European influence, it is perhaps not too far of a stretch to conclude that Japan made the right decision in pursuing a path of relative isolationism. As history unfolded during the next 400 years, in general, countries that embraced European hegemony, whether by choice or by force, tended to suffer from pernicious wealth inequality, perennial political instability, and protracted underdevelopment.

5. It can best be inferred from the passage that in 1636, the Japanese government:
A. Saw its citizens living abroad as potential threats
B. Considered all foreign religions a danger
C. Disagreed with the European philosophy that trade brought wealth
D. Foresaw the economic dangers of European trade and imperialism
E. Believed that ideas coming into Japan via foreign interactions provided no positive impact to Japanese society

6. Which of the following best characterizes the most significant motivation for Europe's behavior with Japan during the 17th century?
A. Religious zeal
B. Long-term political concerns
C. Short-term economic self-interest

D. Cultural imperialism
E. Territorial aggrandizemen

7. The author most likely included the quotation from Article Eight of the Edict at the end of the second paragraph to:
A. Highlight the venomous anger many Japanese leaders felt toward the importation of foreign religions
B. Emphasize the determination of the Japanese government to protect itself from foreign influences it saw as damaging
C. Illustrate how pervasive foreign religious influence had become in Japanese society
D. Emphasize that European economic influence offered no justification for the Edict and the government relied instead on foreign religious influence to justify the Edict
E. Provide an example of Japan's effort to curb cultural and economic exchange

8. Based upon the passage, the author would likely agree most strongly with which of the following statements:
A. European decisions made during the 16th and 17th centuries in dealing with Japan represent an aberration from the typical pattern of European decisions
B. Japanese rulers who responded with ferocity to European influence bear part of the responsibility for the caustic European-Japanese relationship that ensued
C. With the hindsight of history, Japan likely made the appropriate decision in extricating itself from European influence
D. European religious and cultural values conflicted with European economic behavior toward Japan
E. The width and breadth of Japan's cultural fabric suffered from its seclusionist policies

9. According to the passage, which of the following constituted the biggest reason for the Seclusion Edict of 1636?
A. Japanese economic potential would be hampered in the long-term
B. European trade amounted to a disproportionate transfer of wealth
C. With growing European influence, the potential for European military action against the Japanese government became too large
D. Traditional Japanese culture and way of life were threatened by European influence
E. Japanese rulers feared the arrival of additional traders and cultural imperialists

10. According to the passage, the Japanese government took all of the following actions in an attempt to protect Japanese culture and way of life EXCEPT:

A. Prohibit Japanese from visiting other countries, even to see family
B. Execute Japanese citizens who settled in other countries but later decided to return to Japan
C. Heavily regulate foreign economic trade
D. Destroy all remnants of Christianity
E. Prohibit criticism of the feudal shogun system of government

11. The primary purpose of the passage is to:
A. Explain the actions of the Tokugawa government of Japan
B. Compare the results of countries that pursued protectionism with those that pursued globalization
C. Explore the consequences of some European trade and exploration along with analyzing a country's response to it
D. Argue for the success of European trade as a means to create wealth and exert influence
E. Elucidate the root of frustration with European imperialism

Decide which sentence is the most preferable in respect to grammar/ meaning and usage suitable for formal writing.

12. A. If man had not evolved a host of different systems of communication; social life would not have been possible.
B. If man has not evolved a host of different systems of communication; social life would not have been possible.
C. If man did not evolve a host of different systems of communication; social life would not have been possible.
D. If man had not evolved a host of different systems of communication; social life would not have possible.
E. None of the above is correct.

13. A. If all the trees had vanished from the face of Earth, life would come to a standstill.
B. If all the trees will have vanished from the face of Earth, life would come to a standstill.
C. If all the trees vanished from the face of Earth, life would come to a standstill.
D. If all the trees vanished from the face of Earth, life would have come to a standstill.
E. None of the above is correct.

14. A. If I were a farmer, I had use the best agricultural practices.
B. If I were a farmer, I would use the best agricultural practices
C. If I were a farmer, I had used the best agricultural practices
D. If I were a farmer, I will use the best agricultural practices
E. None of the above is correct.

15. A. You get fat unless you eat junk food frequently.
B. You get fat if you eat junk food frequently.
C. You get fat if you ate junk food frequently.
D. You get fat unless you ate junk food frequently.
E. None of the above is correct.

16. A. If the sun sets in the East, everyone would be dumbstruck.
B. If the sun would set in the East, everyone would be dumbstruck.
C. If the sun set in the East, everyone would be dumbstruck.
D. If the sun will set in the East, everyone would be dumbstruck.
E. None of the above is correct.

17. A. If you do not perform well in the exams, you will not be allowed to take guitar classes.
B. If you do not perform well in the exams, you will not be allow to take guitar classes.
C. If you do not perform well in the exams, you be not allowed to take guitar classes.
D. If you do not perform well in the exams, you will not allowed to take guitar classes.
E. None of the above is correct.

18. A. If you have treated me fair, I would have been your friend for a lifetime.
B. If you treated me fair, I would have been your friend for a lifetime.
C. If you will treat me fair, I would have been your friend for a lifetime.
D. If you had treated me fair, I would have been your friend for a lifetime.
E. None of the above is correct.

19. A. If Harry Potter join me for an adventurous trip, I would be on cloud nine.
B. If Harry Potter will join me for an adventurous trip, I would be on cloud nine.

C. If Harry Potter joined me for an adventurous trip, I would be on cloud nine.

D. If Harry Potter joined me for an adventurous trip, I would have been on cloud nine.

E. None of the above is correct.

20. A. If you maintain good oral hygiene, your teeth would remain in good condition.

B. If you maintain good oral hygiene, your teeth would remained in good condition.

C. If you had maintained good oral hygiene, your teeth would remain in good condition.

D. If you maintain good oral hygiene, your teeth will remain in good condition.

E. None of the above is correct.

21. A. The computer get infected with virus, if you do not install antivirus in it.

B. The computer get infected with virus, if you do not installed antivirus in it.

C. The computer will get infected with virus, if you did not install antivirus in it.

D. The computer had got infected with virus. if you do not install antivirus in it.

E. None of the above is correct.

22. A. Please put on the red dress instead of the blue one.

B. Please placed on the red dress instead of the blue one.

C. Please put up the red dress instead of the blue one.

D. Please put in the red dress instead of the blue one.

E. None of the above is correct.

23. A. The prisoners were taken off by the police.

B. The prisoners were taken on by the police.

C. The prisoners were taken away by the police.

D. The prisoners were taken up by the police.

E. None of the above is correct.

24. A. Jim was tired of digging, so his friend helped him over.

B. Jim was tired of digging, so his friend helped him in.

C. Jim was tired of digging, so his friend helped him out.
D. Jim was tired of digging, so his friend helped him on.
E. None of the above is correct.

25. A. You take on your grandfather. Both of you have the same nose and mouth.
B. You take about your grandfather. Both of you have the same nose and mouth.
C. You take of your grandfather. Both of you have the same nose and mouth.
D. You take after your grandfather. Both of you have the same nose and mouth.
E. None of the above is correct.

26. A. Pull over at the next petrol station. I have to go to the washroom.
B. Pull on at the next petrol station. I have to go to the washroom.
C. Pull out at the next petrol station. I have to go to the washroom.
D. Pull in at the next petrol station. I have to go to the washroom.
E. None of the above is correct.

27. A. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to throw in.
B. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to pitch in.
C. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to give in.
D. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to contribute in.
E. None of the above is correct.

28. A. My father and I don't get around. In fact, we always argue.
B. My father and I don't get up. In fact, we always argue.
C. My father and I don't get aside. In fact, we always argue.
D. My father and I don't get along. In fact, we always argue.
E. None of the above is correct.

29. A. I tried calling her all night, but I couldn't get on.
B. I tried calling her all night, but I couldn't get by.
C. I tried calling her all night, but I couldn't get into.

D. I tried calling her all night, but I couldn't get through.
E. None of the above is correct.

30. A. Were they able to put on the fire?
B. Were they able to put out the fire?
C. Were they able to put away the fire?
D. Were they able to put of the fire?
E. None of the above is correct.

31. A. You will get over the illness more quickly if you relax.
B. You will get at the illness more quickly if you relax.
C. You will get up the illness more quickly if you relax.
D. You will get with the illness more quickly if you relax.
E. None of the above is correct.

32. A. I don't know why this class is always so dirty. I get it cleaned every morning.
B. I don't know why this class is always so dirty. I got it cleaned every morning.
C. I don't know why this class is always so dirty. I get it clean every morning.
D. I don't know why this class is always so dirty. I got it clean every morning.
E. None of the above is correct.

33. A. I had to have my digital camera fixing after I dropped it in the water.
B. I had to have my digital camera fixed after I dropped it in the water.
C. I had to have my digital camera to fix after I dropped it in the water.
D. I had to have my digital camera fix after I dropped it in the water.
E. None of the above is correct.

34. A. I can't believe they allow their dog slept on their bed.
B. I can't believe they allow their dog to sleep on their bed.
C. I can't believe they allow their dog sleep on their bed.
D. I can't believe they allow their dog to slept on their bed.
E. None of the above is correct.

35. A. The regulations require you paid your full tuition before classes start.
B. The regulations require you paying your full tuition before classes start.
C. The regulations require you to pay your full tuition before classes start.
D. The regulations require you pay your full tuition before classes start.

E. None of the above is correct.

36. A. Where did you get your car repair? We need some work done on our Toyota and we're looking for a good mechanic.
B. Where did you get your car to repair? We need some work done on our Toyota and we're looking for a good mechanic.
C. Where did you get your car repaired? We need some work done on our Toyota and we're looking for a good mechanic.
D. Where did you get your car to repaired? We need some work done on our Toyota and we're looking for a good mechanic.
E. None of the above is correct.

Identify the part that contains error.

37. Professor Singh had (A)/ her students read four short novels (B)/ in one week. She also made them reading (C)/ five plays in one week. (D)/ No Error (E)
38. The teacher had (A)/ everyone wrote a (B)/ story about a (C)/ special childhood experience. (D)/ No Error (E)
39. Why don't you (A)/ let me to cook (B)/ tonight? You may (C)/ take some rest. (D)/ No Error (E)
40. They are having (A)/ their daughter (B)/ teach by an old (C)/ French woman. (D)/ No Error (E)
41. I went to the hairdresser (A)/ to have cut my hair. (B)/ The hairdresser did (C)/ a pretty good job. (D)/ No Error (E)
42. India has never (A)/ lost a match to (B)/ Pakistan in the world cup. (C)/ Twenty four matches have been played among them. (D)/ No Error (E)
43. India has never (A)/ won a one day from (B)/ 1991to England (C)/ on English soil. (D)/ No Error (E)
44. He studied B. Tech at GLA University (A)/ in Mathura (B)/ since 2014 (C)/ to 2018. (D)/ No Error (E)
45. I asked him to meet me (A)/ on the station. (B)/ He kept waiting under the shed (C)/ beside the station. (D)/ No Error (E)
46. I met your friend (A)/at the cultural event. (B)/ He was at a hurry (C)/ and looking for you. (D)/ No Error (E)
47. By the time he arrived (A)/ the event was over. (B)/ He went off (C)/ without meeting you. (D)/ No Error (E)

48. We walked (A)/ to miles before (B)/ we could get (C)/ the bus for Mathura. (D)/ No Error (E)
49. Trains to London (A)/ leave on time every hour. (B)/ I am impressed (C)/ with the service. (D)/ No Error (E)
50. Why don't you get (A)/ in the train? (B)/ It is going to depart soon. (C)/ You will miss it. (D)/ No Error (E)
51. He wrote the letter to his father (A)/ by the help (B)/ of his friend (C)/ with a blue pen. (D)/ No Error (E)
52. There were a great many question (A) / which were (B) / left unanswered (C) / at the end of the enquiry. (D) / No Error (E)
53. Agriculture is the backbone of the (A)/ Indian economy but the sad part (B)/ was that it is still in a rudimentary stage, (C)/ which ultimately affects the overall economy. (D)/ No Error (E)
54. The proposed INO project, (A) /primarily aims to study atmospheric neutrinos (B)/ in a 1300 m deep cavern in (C)/ the Bodi West Hills in Tamil Nadu. (D) /No Error (E)
55. You cannot (A)/ master physics (B)/ without learning the (C)/ formula that govern it. (D) / No Error (E)
56. The rail coach has been (A)/ shifted near the station (B)/ and would be developed (C)/ as a unique restaurant. (D)/ No Error (E)
57. A shoal (A) / of wolves (B) / attacked (C) / the travelers. (D)/ No Error (E)
58. The Government cannot (A)/ be able to control (B)/ Encephalitis by spending (C)/ just 1% of GDP on Health sector. (D)/ No Error (E)
59. Heavy Hails in many (A)/ parts of apple growing (B)/ areas also resulted in (C)/ huge losses to farmers. (D)/ No Error (E)
60. Just to the North of India (A) / are the Himalayas (B) / that were once (C)/ impregnable. (D) / No Error (E)
61. The price of (A)/ production in the (B)/ industrial sector (C)/ has gone up. (D)/ No Error (E)
62. The murderer was (A)/ convicted (B)/ and ordered to (C)/ be hung. (D) / No Error (E)
63. The central government (A) / is all set to issue (B)/ an order changing the (C)/ very nature of students. (D)/ No Error (E)
64. Myself and Sumit (A) / will take care (B)/ of the function (C)/ on Tuesday. (D) / No Error (E)
65. The company has (A)/ appointed consultants (B)/ to help them (C) / increase its revenue. (D)/ No Error (E)

66. Women after women (A) / spoke against (B) / the practice (C) / of dowry. (D) / No Error (E)

67. While driving (A)/ along the narrow village roads (B)/ in his beat-up Land Rover, (C)/ a great idea struck him.(D)/ No Error (E)

68. Ranging from mild to acute, (A)/ UNICEF reports that (B)/ 20 per cent of Iraq's 3.5 million children (C)/ suffer from malnutrition. (D)/ No Error (E)

69. Many readers objected (A)/ to the newspaper's picture (B)/ of a young man posed (C)/ beside a dead whale (D)/ wearing running shoes and holding a gun. (E)

70. Photovoltaic cells convert (A)/ light energy from the sun (B)/ and store this energy (C)/ in batteries for easy use. (D)/ No Error (E)

Find out Correct Sequence of paragraph:

71. A. Such a man goes on working hard and even if he fails he is never downcast.
B. It is therefore, the man who labours hard with a strong resolution and an unshaken will, who achieves success and makes his fortune.
C. In turn failures make him all the more determined and resolute and he persists in his task till he attains the desired success.
D. A man who possesses a strong will and firm determination finds all difficulties solved.
E. To him there are a thousand ways open to steer clear of all dangers and difficulties.
A. DEACB B. ABCDE C. DBACE D. AEBCD E. None

72. A. With the passage of time, vices become more apparent and virtues become objects of jealousy and envy, thereby causing contempt and hatred in the hearts of each other.
B. They become familiar with not only strengths but also weaknesses of each other's characters.
C. Generally people think that familiarity should breed love, mutual understanding and tolerance.
D. They expect that coming together of two persons should bring them closer and forge the bond of kinship between them.
E. But when two persons come closer, they come to know not only strengths but also weaknesses of each other's character.

A. ABCDE B. CDEBA C. CABED D. DBCAE E. None

73. A. He used to go to his school and enjoyed studying and mid day meals there.
B. Besides, they needed his help in odd jobs in the house as well as their fields.
C. Karma lived with his parents in a thatched small house in a silent corner of the village.
D. This all stopped as his parents could not afford to send him to school to impart further education to him.
A. BDAC B. ABCD C. CADB D. CBAD E. None

74. A. But sometimes, the persons of opposite nature also come closer fall in each other's company by accident, chance or out of ignorance vitiating the above statement to some extent.
B. If a man moves in the company of good, gentle and noble people, he is usually adjudged to be a gentleman.
C. It is usual for a man to see company of those who possess tastes, tendencies and temperaments like his own.
D. On the other side, if he keeps company with evil persons and bad characters, he is considered to be a man of bad character.
E. Generally, the character and conduct of a person is gauged by the kinds of people he mixes and moves with.
A. ABCDE B. DECAB C. BDACE D. EBDCA E. None

75. A. He cannot achieve wealth and fortune unless he works hard and unless he has a powerful will.
B. It follows from it that he is the builder of his own fortune.
C. A person must be persevering and must exert himself if he wants to shape out his destiny.
D. He can also bring misery and sufferings upon himself if he has not the wish to labour.
E. He can make his fortune if he wishes to do so by being industrious and persevering.
A. CABED B. ABCDE C. EBACD D. CBAED E. None

76. A. When finally they made their first ascent from the desert tract beside the sea, to be borne aloft for almost a whole minute, a great change was effected in the nation's attitude.

B. Accordingly, the brothers, each a man of mettle and each the perfect complement to the other, set out with their ingenious device, but with their very little capital.
C. Those who had formerly been skeptical and had prophesied that the Wright machine would remain forever stationary on the earth, were loudest in their praise of the pioneers of the air.
D. Men laughed at the Wright Brothers, mechanics from Dayton, saying that a practicable flying machine would never be built and counselled them to stay on the ground.
E. The wrights, however, refused to accept this advice or to alter their plans, for they were certain that their machine embodied the principles of aviation and they were confident of their success.
A. DACBE B. ABCDE C. DEBAC D. BDEAC E. None

77. A. The belief in evil spirits, though still common, is probably less wide spread than it was.
B. They live in terror of germs and practise elaborate antiseptic rites in order to counteract their influence.
C. Evil spirits being out of fashion, it must therefore find expression in other beliefs.
D. With many people especially women, bacilli have taken the place of spirits and microbes for them are personification of evil.
E. But the human tendency to oversize its sense of value is still as strong as ever.
F. The forms change but the substance remains.
A. AEBDCF B. ABCDEF C. BCADEF D. CEFABD
E. None

78. A. Special justice has always exercised an appeal to sensitive persons.
B. We talk of things material and spiritual.
C. Every great wave of human thought which has affected millions of human beings has something spiritual in it.
D. Even the great revolution would not have succeeded without a spiritual element.
E. Yet it is a little difficult to draw a line between the two.
A. BADEC B. BECDA C. ABCDE D. CDEAB E. None

79. A. Then we come to classical Sanskrit which is also very good.
B. Instead of being inspired by good idea, we have even lost what we had.
C. However it gradually deteriorates and there is no vitality left it in it.

D. We start with magnificent literature.
E. The process of decay through centuries can be traced in our literature.
A. EDACB B. BACDE C. EACBD D. CDABE E. None

80. A. It must be viewed, as some new epidemic would be viewed, as a common peril to be met by concerted action.
B. If we are to think wisely about the new problems raised by nuclear weapons, we must learn to view the whole matter in a quite different way.
C. These conflicts are so virulent and so passionate that they produce a wide spread inability to understand even very obvious matters.
D. It is a profound misfortune that the whole question of nuclear warfare has become entangled in the age-old conflicts of power politics.
A. ABCD B. DCBA C. DBAC D. CDAB E. None

Choose the correct idiom/word that explains the given expression.

81. A meeting format designed to accelerate business contacts.
A. Put in the picture B. Speed Networking C. Utopia
D. Hit the air waves E. Pedestrian

82. The world in miniature.
A. Stool Pigeon B. Mimic C. Microcosm D. Niche
E. Drop a line

83. A clever or expert way of doing things, especially in a job.
A. Tricks of the trade B. Get the axe C. Plum job
D. Pound the pavement E. Pastiche

84. One who compiles a dictionary.
A. Linguist B. Lexicographer C. Anthropologist D. Archeologist
E. Laureate

85. Be discharged or fired from the job.
A. Infantry B. Lunatic C. Get the axe D. Plum job
E. Golden handcuffs

86. A study of the origin of words.
A. Syntax B. Semantics C. Etymology D. Linguistics
E. Numismatics

87. A desirable position which is well-paid and considered relatively easy.
A. Get the axe B. Plum job C. Golden handcuffs
D. Duty bound E. Espionage

88. One who easily believes what others say.

A. Credulous B. Complacent C. Congenital D. Delegate
E. Equivocal

89. A government by the few.
A. Plutocracy B. Anarchy C. Oligarchy D. Autocracy
E. Bureaucracy

90. A great lover of books.
A. Betrothed B. Bibliophile C. Centenarian D. Misanthrope
E. Omnivorous

SET-B

University Roll No._____

End Term Examination, 2018-19

B. Tech. 2nd Year (Even Semester)

Subject: English for Professional Purposes II

Subject Code: BELH0004

Time: 100 Minutes

Max. Marks: 30

VERBAL APTITUDE

(Answer all the questions)

Choose the correct idiom/word that explains the given expression.

1. A meeting format designed to accelerate business contacts.
   A. Put in the picture B. Speed Networking C. Utopia
   D. Hit the air waves E. Pedestrian
2. The world in miniature.
   A. Stool Pigeon B. Mimic C. Microcosm D. Niche
   E. Drop a line
3. A clever or expert way of doing things, especially in a job.
   A. Tricks of the trade B. Get the axe C. Plum job
   D. Pound the pavement E. Pastiche
4. One who compiles a dictionary.
   A. Linguist B. Lexicographer C. Anthropologist D. Archeologist
   E. Laureate
5. Be discharged or fired from the job.
   A. Infantry B. Lunatic C. Get the axe D. Plum job
   E. Golden handcuffs
6. A study of the origin of words.
   A. Syntax B. Semantics C. Etymology D. Linguistics
   E. Numismatics
7. A desirable position which is well-paid and considered relatively easy.
   A. Get the axe B. Plum job C. Golden handcuffs
   D. Duty bound E. Espionage
8. One who easily believes what others say.
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A. Plutocracy B. Anarchy C. Oligarchy D. Autocracy
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10. A great lover of books.
A. Betrothed B. Bibliophile C. Centenarian D. Misanthrope
E. Omnivorous

Identify the part that contains error.

11. Professor Singh had (A)/ her students read four short novels (B)/ in one week. She also made them reading (C)/ five plays in one week. (D)/ No Error (E)
12. The teacher had (A)/ everyone wrote a (B)/ story about a (C)/ special childhood experience. (D)/ No Error (E)
13. Why don't you (A)/ let me to cook (B)/ tonight? You may (C)/ take some rest. (D)/ No Error (E)
14. They are having (A)/ their daughter (B)/ teach by an old (C)/ French woman. (D)/ No Error (E)
15. I went to the hairdresser (A)/ to have cut my hair. (B)/ The hairdresser did (C)/ a pretty good job. (D)/ No Error (E)
16. India has never (A)/ lost a match to (B)/ Pakistan in the world cup. (C)/ Twenty four matches have been played among them. (D)/ No Error (E)
17. India has never (A)/ won a one day from (B)/ 1991to England (C)/ on English soil. (D)/ No Error (E)
18. He studied B. Tech at GLA University (A)/ in Mathura (B)/ since 2014 (C)/ to 2018. (D)/ No Error (E)
19. I asked him to meet me (A)/ on the station. (B)/ He kept waiting under the shed (C)/ beside the station. (D)/ No Error (E)
20. I met your friend (A)/at the cultural event. (B)/ He was at a hurry (C)/ and looking for you. (D)/ No Error (E)
21. By the time he arrived (A)/ the event was over. (B)/ He went off (C)/ without meeting you. (D)/ No Error (E)
22. We walked (A)/ to miles before (B)/ we could get (C)/ the bus for Mathura. (D)/ No Error (E)
23. Trains to London (A)/ leave on time every hour. (B)/ I am impressed (C)/ with the service. (D)/ No Error (E)
24. Why don't you get (A)/ in the train? (B)/ It is going to depart soon. (C)/ You will miss it. (D)/ No Error (E)

25. He wrote the letter to his father (A)/ by the help (B)/ of his friend (C)/ with a blue pen. (D)/ No Error (E)

26. There were a great many question (A) / which were (B) / left unanswered (C) / at the end of the enquiry. (D) / No Error (E)

27. Agriculture is the backbone of the (A)/ Indian economy but the sad part (B)/ was that it is still in a rudimentary stage, (C)/ which ultimately affects the overall economy. (D)/ No Error (E)

28. The proposed INO project, (A) /primarily aims to study atmospheric neutrinos (B)/ in a 1300 m deep cavern in (C)/ the Bodi West Hills in Tamil Nadu. (D) /No Error (E)

29. You cannot (A)/ master physics (B)/ without learning the (C)/ formula that govern it. (D) / No Error (E)

30. The rail coach has been (A)/ shifted near the station (B)/ and would be developed (C)/ as a unique restaurant. (D)/ No Error (E)

31. A shoal (A) / of wolves (B) / attacked (C) / the travelers. (D)/ No Error (E)

32. The Government cannot (A)/ be able to control (B)/ Encephalitis by spending (C)/ just 1% of GDP on Health sector. (D)/ No Error (E)

33. Heavy Hails in many (A)/ parts of apple growing (B)/ areas also resulted in (C)/ huge losses to farmers. (D)/ No Error (E)

34. Just to the North of India (A) / are the Himalayas (B) / that were once (C)/ impregnable. (D) / No Error (E)

35. The price of (A)/ production in the (B)/ industrial sector (C)/ has gone up. (D)/ No Error (E)

36. The murderer was (A)/ convicted (B)/ and ordered to (C)/ be hung. (D) / No Error (E)

37. The central government (A) / is all set to issue (B)/ an order changing the (C)/ very nature of students. (D)/ No Error (E)

38. Myself and Sumit (A) / will take care (B)/ of the function (C)/ on Tuesday. (D) / No Error (E)

39. The company has (A)/ appointed consultants (B)/ to help them (C) / increase its revenue. (D)/ No Error (E)

40. Women after women (A) / spoke against (B) / the practice (C) / of dowry. (D) / No Error (E)

41. While driving (A)/ along the narrow village roads (B)/ in his beat-up Land Rover, (C)/ a great idea struck him.(D)/ No Error (E)

42. Ranging from mild to acute, (A)/ UNICEF reports that (B)/ 20 per cent of Iraq's 3.5 million children (C)/ suffer from malnutrition. (D)/ No Error (E)

43. Many readers objected (A)/ to the newspaper's picture (B)/ of a young man posed (C)/ beside a dead whale (D)/ wearing running shoes and holding a gun. (E)

44. Photovoltaic cells convert (A)/ light energy from the sun (B)/ and store this energy (C)/ in batteries for easy use. (D)/ No Error (E)

Find out Correct Sequence of paragraph:

45. A. Such a man goes on working hard and even if he fails he is never downcast.
B. It is therefore, the man who labours hard with a strong resolution and an unshaken will, who achieves success and makes his fortune.
C. In turn failures make him all the more determined and resolute and he persists in his task till he attains the desired success.
D. A man who possesses a strong will and firm determination finds all difficulties solved.
E. To him there are a thousand ways open to steer clear of all dangers and difficulties.
A. DEACB B. ABCDE C. DBACE D. AEBCD E. None

46. A. With the passage of time, vices become more apparent and virtues become objects of jealousy and envy, thereby causing contempt and hatred in the hearts of each other.
B. They become familiar with not only strengths but also weaknesses of each other's characters.
C. Generally people think that familiarity should breed love, mutual understanding and tolerance.
D. They expect that coming together of two persons should bring them closer and forge the bond of kinship between them.
E. But when two persons come closer, they come to know not only strengths but also weaknesses of each other's character.
A. ABCDE B. CDEBA C. CABED D. DBCAE E. None

47. A. He used to go to his school and enjoyed studying and mid day meals there.
B. Besides, they needed his help in odd jobs in the house as well as their fields.
C. Karma lived with his parents in a thatched small house in a silent corner of the village.

D. This all stopped as his parents could not afford to send him to school to impart further education to him.

A. BDAC B. ABCD C. CADB D. CBAD E. None

48. A. But sometimes, the persons of opposite nature also come closer fall in each other's company by accident, chance or out of ignorance vitiating the above statement to some extent.

B. If a man moves in the company of good, gentle and noble people, he is usually adjudged to be a gentleman.

C. It is usual for a man to see company of those who possess tastes, tendencies and temperaments like his own.

D. On the other side, if he keeps company with evil persons and bad characters, he is considered to be a man of bad character.

E. Generally, the character and conduct of a person is gauged by the kinds of people he mixes and moves with.

A. ABCDE B. DECAB C. BDACE D. EBDCA E. None

49. A. He cannot achieve wealth and fortune unless he works hard and unless he has a powerful will.

B. It follows from it that he is the builder of his own fortune.

C. A person must be persevering and must exert himself if he wants to shape out his destiny.

D. He can also bring misery and sufferings upon himself if he has not the wish to labour.

E. He can make his fortune if he wishes to do so by being industrious and persevering.

A. CABED B. ABCDE C. EBACD D. CBAED E. None

50. A. When finally they made their first ascent from the desert tract beside the sea, to be borne aloft for almost a whole minute, a great change was effected in the nation's attitude.

B. Accordingly, the brothers, each a man of mettle and each the perfect complement to the other, set out with their ingenious device, but with their very little capital.

C. Those who had formerly been skeptical and had prophesied that the Wright machine would remain forever stationary on the earth, were loudest in their praise of the pioneers of the air.

D. Men laughed at the Wright Brothers, mechanics from Dayton, saying that a practicable flying machine would never be built and counselled them to stay on the ground.

E. The wrights, however, refused to accept this advice or to alter their plans, for they were certain that their machine embodied the principles of aviation and they were confident of their success.
A. DACBE B. ABCDE C. DEBAC D. BDEAC E. None

51. A. The belief in evil spirits, though still common, is probably less wide spread than it was.
B. They live in terror of germs and practise elaborate antiseptic rites in order to counteract their influence.
C. Evil spirits being out of fashion, it must therefore find expression in other beliefs.
D. With many people especially women, bacilli have taken the place of spirits and microbes for them are personification of evil.
E. But the human tendency to oversize its sense of value is still as strong as ever.
F. The forms change but the substance remains.
A. AEBDCF B. ABCDEF C. BCADEF D. CEFABD
E. None

52. A. Special justice has always exercised an appeal to sensitive persons.
B. We talk of things material and spiritual.
C. Every great wave of human thought which has affected millions of human beings has something spiritual in it.
D. Even the great revolution would not have succeeded without a spiritual element.
E. Yet it is a little difficult to draw a line between the two.
A. BADEC B. BECDA C. ABCDE D. CDEAB E. None

53. A. Then we come to classical Sanskrit which is also very good.
B. Instead of being inspired by good idea, we have even lost what we had.
C. However it gradually deteriorates and there is no vitality left it in it.
D. We start with magnificent literature.
E. The process of decay through centuries can be traced in our literature.
A. EDACB B. BACDE C. EACBD D. CDABE E. None

54. A. It must be viewed, as some new epidemic would be viewed, as a common peril to be met by concerted action.
B. If we are to think wisely about the new problems raised by nuclear weapons, we must learn to view the whole matter in a quite different way.

C. These conflicts are so virulent and so passionate that they produce a wide spread inability to understand even very obvious matters.
D. It is a profound misfortune that the whole question of nuclear warfare has become entangled in the age-old conflicts of power politics.
A. ABCD B. DCBA C. DBAC D. CDAB E. None

Decide which sentence is the most preferable in respect to grammar/ meaning and usage suitable for formal writing.

55. A. If man had not evolved a host of different systems of communication; social life would not have been possible.
B. If man has not evolved a host of different systems of communication; social life would not have been possible.
C. If man did not evolve a host of different systems of communication; social life would not have been possible.
D. If man had not evolved a host of different systems of communication; social life would not have possible.
E. None of the above is correct.

56. A. If all the trees had vanished from the face of Earth, life would come to a standstill.
B. If all the trees will have vanished from the face of Earth, life would come to a standstill.
C. If all the trees vanished from the face of Earth, life would come to a standstill.
D. If all the trees vanished from the face of Earth, life would have come to a standstill.
E. None of the above is correct:

57. A. If I were a farmer, I had use the best agricultural practices.
B. If I were a farmer, I would use the best agricultural practices
C. If I were a farmer, I had used the best agricultural practices
D. If I were a farmer, I will use the best agricultural practices
E. None of the above is correct.

58. A. You get fat unless you eat junk food frequently.
B. You get fat if you eat junk food frequently.
C. You get fat if you ate junk food frequently.
D. You get fat unless you ate junk food frequently.

E. None of the above is correct.

59. A. If the sun sets in the East, everyone would be dumbstruck.
B. If the sun would set in the East, everyone would be dumbstruck.
C. If the sun set in the East, everyone would be dumbstruck.
D. If the sun will set in the East, everyone would be dumbstruck.
E. None of the above is correct.

60. A. If you do not perform well in the exams, you will not be allowed to take guitar classes.
B. If you do not perform well in the exams, you will not be allow to take guitar classes.
C. If you do not perform well in the exams, you be not allowed to take guitar classes.
D. If you do not perform well in the exams, you will not allowed to take guitar classes.
E. None of the above is correct.

61. A. If you have treated me fair, I would have been your friend for a lifetime.
B. If you treated me fair, I would have been your friend for a lifetime.
C. If you will treat me fair, I would have been your friend for a lifetime.
D. If you had treated me fair, I would have been your friend for a lifetime.
E. None of the above is correct.

62. A. If Harry Potter join me for an adventurous trip, I would be on cloud nine.
B. If Harry Potter will join me for an adventurous trip, I would be on cloud nine.
C. If Harry Potter joined me for an adventurous trip, I would be on cloud nine.
D. If Harry Potter joined me for an adventurous trip, I would have been on cloud nine.
E. None of the above is correct.

63. A. If you maintain good oral hygiene, your teeth would remain in good condition.
B. If you maintain good oral hygiene, your teeth would remained in good condition.

C. If you had maintained good oral hygiene, your teeth would remain in good condition.
D. If you maintain good oral hygiene, your teeth will remain in good condition.
E. None of the above is correct.

64. A. The computer get infected with virus, if you do not install antivirus in it.
B. The computer get infected with virus, if you do not installed antivirus in it.
C. The computer will get infected with virus, if you did not install antivirus in it.
D. The computer had got infected with virus, if you do not install antivirus in it.
E. None of the above is correct.

65. A. Please put on the red dress instead of the blue one.
B. Please placed on the red dress instead of the blue one.
C. Please put up the red dress instead of the blue one.
D. Please put in the red dress instead of the blue one.
E. None of the above is correct.

66. A. The prisoners were taken off by the police.
B. The prisoners were taken on by the police.
C. The prisoners were taken away by the police.
D. The prisoners were taken up by the police.
E. None of the above is correct.

67. A. Jim was tired of digging, so his friend helped him over.
B. Jim was tired of digging, so his friend helped him in.
C. Jim was tired of digging, so his friend helped him out.
D. Jim was tired of digging, so his friend helped him on.
E. None of the above is correct.

68. A. You take on your grandfather. Both of you have the same nose and mouth.
B. You take about your grandfather. Both of you have the same nose and mouth.
C. You take of your grandfather. Both of you have the same nose and mouth.

D. You take after your grandfather. Both of you have the same nose and mouth.
E. None of the above is correct.

69. A. Pull over at the next petrol station. I have to go to the washroom.
B. Pull on at the next petrol station. I have to go to the washroom.
C. Pull out at the next petrol station. I have to go to the washroom.
D. Pull in at the next petrol station. I have to go to the washroom.
E. None of the above is correct.

70. A. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to throw in.
B. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to pitch in.
C. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to give in.
D. Person 1: Does everyone have to contribute?
Person 2: Yes, everyone has to contribute in.
E. None of the above is correct.

71. A. My father and I don't get around. In fact, we always argue.
B. My father and I don't get up. In fact, we always argue.
C. My father and I don't get aside. In fact, we always argue.
D. My father and I don't get along. In fact, we always argue.
E. None of the above is correct.

72. A. I tried calling her all night, but I couldn't get on.
B. I tried calling her all night, but I couldn't get by.
C. I tried calling her all night, but I couldn't get into.
D. I tried calling her all night, but I couldn't get through.
E. None of the above is correct.

73. A. Were they able to put on the fire?
B. Were they able to put out the fire?
C. Were they able to put away the fire?
D. Were they able to put of the fire?
E. None of the above is correct.

74. A. You will get over the illness more quickly if you relax.
B. You will get at the illness more quickly if you relax.

C. You will get up the illness more quickly if you relax.
D. You will get with the illness more quickly if you relax.
E. None of the above is correct.

75. A. I don't know why this class is always so dirty. I get it cleaned every morning.
B. I don't know why this class is always so dirty. I got it cleaned every morning.
C. I don't know why this class is always so dirty. I get it clean every morning.
D. I don't know why this class is always so dirty. I got it clean every morning.
E. None of the above is correct.

76. A. I had to have my digital camera fixing after I dropped it in the water.
B. I had to have my digital camera fixed after I dropped it in the water.
C. I had to have my digital camera to fix after I dropped it in the water.
D. I had to have my digital camera fix after I dropped it in the water.
E. None of the above is correct.

77. A. I can't believe they allow their dog slept on their bed.
B. I can't believe they allow their dog to sleep on their bed.
C. I can't believe they allow their dog sleep on their bed.
D. I can't believe they allow their dog to slept on their bed.
E. None of the above is correct.

78. A. The regulations require you paid your full tuition before classes start.
B. The regulations require you paying your full tuition before classes start.
C. The regulations require you to pay your full tuition before classes start.
D. The regulations require you pay your full tuition before classes start.

E. None of the above is correct.

79. A. Where did you get your car repair? We need some work done on our Toyota and we're looking for a good mechanic.
B. Where did you get your car to repair? We need some work done on our Toyota and we're looking for a good mechanic.
C. Where did you get your car repaired? We need some work done on our Toyota and we're looking for a good mechanic.
D. Where did you get your car to repaired? We need some work done on our Toyota and we're looking for a good mechanic.

E. None of the above is correct.

Read the passages given below and answer the questions that follow.

Passage 1

If a supernova (the explosion of a massive star) triggered star formation from dense clouds of gas and dust, and if the most massive star to be formed from the cloud evolved into a supernova and triggered a new round of star formation, and so on, then a chain of star-forming regions would result. If many such chains were created in a differentially rotating galaxy, the distribution of stars would resemble the observed distribution in a spiral galaxy. This line of reasoning underlies an exciting new theory of spiral-galaxy structure. A computer simulation based on this theory has reproduced the appearance of many spiral galaxies without assuming an underlying density wave, the hallmark of the most widely accepted theory of the large-scale structure of spiral galaxies. That theory maintains that a density wave of spiral form sweeps through the central plane of a galaxy, compressing clouds of gas and dust, which collapse into stars that form a spiral pattern.

80. The primary purpose of the passage is to
A. describe what results when a supernova triggers the creation of chains of star-forming regions
B. propose a modification in the most widely accepted theory of spiral-galaxy structure
C. compare and contrast the roles of clouds of gas and dust in two theories of spiral-galaxy structure
D. describe a new theory of spiral-galaxy structure and contrast it with the most widely accepted theory
E. describe a new theory of spiral-galaxy structure and discuss a reason why it is inferior to the most widely accepted theory

81. The passage implies that, according to the new theory of spiral-galaxy structure, a spiral galaxy can be created by supernovas when the supernovas are
A. producing an underlying density wave
B. affected by a density wave of spiral form
C. distributed in a spiral pattern
D. located in the central plane of a galaxy
E. located in a differentially rotating galaxy

82. Which of the following, if true, would most discredit the new theory as described in the passage?
A. The exact mechanism by which a star becomes a supernova is not yet completely known and may even differ for different stars.
B. Chains of star-forming regions like those postulated in the new theory have been observed in the vicinity of dense clouds of gas and dust.
C. The most massive stars formed from supernova explosions are unlikely to evolve into supernovas.
D. Computer simulations of supernovas provide a poor picture of what occurs just before a supernova explosion.
E. A density wave cannot compress clouds of gas and dust to a density high enough to create a star.

83. The author's attitude toward the new theory of spiral-galaxy structure can best be described as
A. euphoric
B. enthusiastic
C. concerned
D. critical
E. disputatious

Passage 2

Although European decisions during the 16th and 17th centuries to explore, trade with, and colonize large portions of the world brought tremendous economic wealth and vast geographic influence, the enormous success of European maritime ventures during the age of exploration also engendered a litany of unintended consequences for most of the nations with which Europe interacted. Due to their incredible military force, religious zeal, and uncompromising goal of profit, Europeans often imposed their traditions, values, and customs on the people with whom they traded. They frequently acted without regard to the long-term welfare of others as their principal concern was short-term economic gain. Since many nations that traded with Europe placed high value on their historical customs, some natives became deeply disconcerted by the changes that occurred as a result of European power. These factors, coupled with perennial domestic political instability, caused numerous countries to grow increasingly resistant to European influence. One potent example of this ideological shift can be seen in the actions of the Tokugawa government of Japan. In its Seclusion Edict of 1636, the government attempted to extricate cultural interactions with Europe from the

intimate fabric of Japanese society. The Edict attempted to accomplish this by focusing on three areas. First, it sought to curb cultural exchange by eliminating people bringing European ideas into Japan. The Edict stated, "Japanese ships shall by no means be sent abroad....All Japanese residing abroad shall be put to death when they return home." Second, the Edict focused on limiting trade. Articles 11 through 17 of the Edict imposed stringent regulations on trade and commerce. Third, the government banned Christianity, which it saw as an import from Europe that challenged the long-established and well-enshrined religious traditions of Japan. The government went to considerable lengths to protect its culture. Article eight of the Edict stated, "Even ships shall not be left untouched in the matter of exterminating Christians." With the example of Japan and the examples of other countries that chose a different response to European influence, it is perhaps not too far of a stretch to conclude that Japan made the right decision in pursuing a path of relative isolationism. As history unfolded during the next 400 years, in general, countries that embraced European hegemony, whether by choice or by force, tended to suffer from pernicious wealth inequality, perennial political instability, and protracted underdevelopment.

84. It can best be inferred from the passage that in 1636, the Japanese government:
A. Saw its citizens living abroad as potential threats
B. Considered all foreign religions a danger
C. Disagreed with the European philosophy that trade brought wealth
D. Foresaw the economic dangers of European trade and imperialism
E. Believed that ideas coming into Japan via foreign interactions provided no positive impact to Japanese society

85. Which of the following best characterizes the most significant motivation for Europe's behavior with Japan during the 17th century?
A. Religious zeal
B. Long-term political concerns
C. Short-term economic self-interest
D. Cultural imperialism
E. Territorial aggrandizemen

86. The author most likely included the quotation from Article Eight of the Edict at the end of the second paragraph to:
A. Highlight the venomous anger many Japanese leaders felt toward the importation of foreign religions
B. Emphasize the determination of the Japanese government to protect itself from foreign influences it saw as damaging

C. Illustrate how pervasive foreign religious influence had become in Japanese society
D. Emphasize that European economic influence offered no justification for the Edict and the government relied instead on foreign religious influence to justify the Edict
E. Provide an example of Japan's effort to curb cultural and economic exchange

87. Based upon the passage, the author would likely agree most strongly with which of the following statements:
A. European decisions made during the 16th and 17th centuries in dealing with Japan represent an aberration from the typical pattern of European decisions
B. Japanese rulers who responded with ferocity to European influence bear part of the responsibility for the caustic European-Japanese relationship that ensued
C. With the hindsight of history, Japan likely made the appropriate decision in extricating itself from European influence
D. European religious and cultural values conflicted with European economic behavior toward Japan
E. The width and breadth of Japan's cultural fabric suffered from its seclusionist policies

88. According to the passage, which of the following constituted the biggest reason for the Seclusion Edict of 1636?
A. Japanese economic potential would be hampered in the long-term
B. European trade amounted to a disproportionate transfer of wealth
C. With growing European influence, the potential for European military action against the Japanese government became too large
D. Traditional Japanese culture and way of life were threatened by European influence
E. Japanese rulers feared the arrival of additional traders and cultural imperialists

89. According to the passage, the Japanese government took all of the following actions in an attempt to protect Japanese culture and way of life EXCEPT:

A. Prohibit Japanese from visiting other countries, even to see family
B. Execute Japanese citizens who settled in other countries but later decided to return to Japan
C. Heavily regulate foreign economic trade
D. Destroy all remnants of Christianity
E. Prohibit criticism of the feudal shogun system of government

90. The primary purpose of the passage is to:
A. Explain the actions of the Tokugawa government of Japan
B. Compare the results of countries that pursued protectionism with those that pursued globalization
C. Explore the consequences of some European trade and exploration along with analyzing a country's response to it
D. Argue for the success of European trade as a means to create wealth and exert influence
E. Elucidate the root of frustration with European imperialism

Paper Code-BTDH/MTDH 0302

SET-A

Soft Skills

End Term, Even Sem Examination (2018-19)

Course- B.tech/MCA Year- II, I

Branch- (All)

Time : 120 Min. M.M-40

Instructions :

1. Attempt all questions.
2. There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.
3. The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.
4. The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.
5. Only black& blue pen should be used to fill forms.
6. Each branch is identified with a specific subject code which should be written in subject column of OMR sheet
7. Write your University roll number in the roll no column of OMR sheet.

Name of Student:-.....
Branch:-.....
Section-.....
Uni. Roll No:.....

All The Best

(Aptitude)-No of question-80, marks-0.50

Q1. Four friends A, B, C, D share Rs. 10,500 in the ratio (1/3): (¼): (1/6): (1/8); how much more money A and C together get than B and D together?
A. 1200 B. 1000 C. 1700 D. 1500
E. None of these

Q2. A, B and C play cricket. A's runs are to B's runs as 3:2 and B's runs are to C's as 3:2. They get altogether 342 runs. How many runs did A make?
A. 162 B. 108 C. 72 D. 64
E. None of these

Q3. Two numbers are respectively 20% and 50% more than a third number. ratio of the two numbers is:
A. 2:5 B. 3:5 C. 4:5 D. 5:4 E. None of these

Q4. A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:
A. 1425 B. 1500 C. 1537.50 D.1476
E. None of these

Q5. A starts business with Rs. 3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?
A. 7500 B. 9000 C. 8000 D.8500
E. None of these

Q6. At what percent per annum will a sum of money double in 16 years?
A. 6.25% B.7.5% C.23/3% D.12.5% E.None of these

Q7. What will be the ratio of simple interest earned by certain amount at the same rate of interest for 6 years and that for 9 years?
A.Data inadequate B.1 : 3 C.2 : 3 D.1 : 4
E.None of these

Q8. The compound interest on a certain sum for 2 years at 10% per annum is Rs. 525. The simple interest on the same sum for double the time at half the rate percent per annum is:
A. Rs. 400 B. Rs. 500 C.Rs. 600 D.Rs. 800
E.None of these

Q9. There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate?
A. Rs2160 B. Rs. 3120 C.Rs. 3972 D.Rs. 6240 E.none of these

Q10. The angle between the minute hand and the hour hand of a clock when the time is 4.20, is:
A. 0° B.10° C.5° D.20° E.None of these

Q11. In 12 hrs, how many times both the hands will overlap?
A.11 B.12 C.22 D.24 E.None of these

Q12. What was the day of the week on 17th June, 1998?
A.Monday B.Tuesday C.Wednesday D.Thursday
E.None of these

Q13. It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?
A.Sunday B.Saturday C.Friday D.Wednesday
E.None of these

Q14. Pointing to a boy, a lady says that his mother is the only sister of her mother. How is the boy related to that lady?
A.Nephew B.Uncle C.Cousin D.Brother E.None of these

Q(15-16)
I. P+Q means P is mother of Q.
II. P-Q means P is sister of Q.
III. P$Q means P is father of Q
IV. P#Q means P is brother of Q

Q15. Which of the following means D is definitely daughter of A?
A. A+B#C#D B. C+A$D-B C. A+C$B-D D. B+A$D#C
E. None of these

Q16. Which of the following means R is brother of T?
A. R+S#U$T B. U+R#S#T C. U+R-S+T D. K#S$Q-R
E. None of these

Q17. A is the brother of K, K is the brother of Q and R is the sister of Q. How is R related to A?
A.Brother B.Sister C.Cousin D.Nephew E.None of these

Q18. 40% of the employees in a factory are workers. All the remaining employees are executives. The annual income of each worker is Rs. 390. The annual income of each executive is Rs. 420. What is the average annual income of all the employees in the factory together?
A. 390 B.405 C.408 D.415 E.None of these

Q19. The average price of 10 books is Rs.12 while the average price of 8 of these books is Rs.11.75. Of the remaining two books, if the price of one book is 60% more than the price of the other, what is the price of each of these two books?
A. Rs5, Rs7.50 B.Rs8, Rs12 C.Rs10, Rs16 D.Rs12, Rs14
E.None of these

Q20. In a set of three numbers, the average of first two numbers is 2, the average of the last two numbers is 3, and the average of the first and the last numbers is 4. What is the average of three numbers?
A.2 B.2.5 C.3 D.3.5 E.None of these

Q21. The average monthly salary of 12 workers and 3 managers in a factory was Rs. 600. When one of the manager whose salary was Rs. 720, was replaced with a new manager, then the average salary of the team went down to 580. What is the salary of the new manager?
A. 570 B.420 C.690 D.640 E.None of these

Q22. Milk and water are in a Can A as 4:1 and in Can B as 3:2. For Can C, if one takes equal quantities from A and B, find the ratio of milk to water in C.
A. 7:3 B. 4:7 C. 3:5 D. 5:4 E. None

Q23. How many litres of water should be added to a 30 litre mixture of milk and water containing milk and water in the ratio of 7 : 3 such that the resultant mixture has 40% water in it?
A. 7 litres B.10 litres C.5 litres D.12 liters E.None of these

Q24.In what ratio must a person mix three kinds of tea costing Rs.60/kg, Rs.75/kg and Rs.100 /kg so that the resultant mixture when sold at Rs.96/kg yields a profit of 20%?
A. 1: 2: 4 B.3: 7: 6 C.1: 4 : 2 D.1:3:2 E.None of these

Q25. If 2 kg metal , of which (1/3) is zinc and the rest is copper , be mixed with 3 kg of metal , of which (1/4) is zinc and the rest is copper . What is the ratio of zinc to copper in the mixture ?
A. 11 : 43 B. 15 : 37 C. 17 : 43 D. 23 : 74 E. 18 : 52

Q26. How many litres of a 12 litre mixture containing milk and water in the ratio of 2: 3 be replaced with pure milk so that the resultant mixture contains milk and water in equal proportion?
A.4 litres B.2 litres C.1 litres D.1.5 litres E.None of these

Q27. THRICE is coded as UJUMHK, then 'EQUALS' will be coded as:
A.FRVMT B.FPVZMU C.FSXEQY D.FRWDPX
E.None of these

Q28. If SCIENTIST is code as QAGCLRGQR, and then what will be the code for GENIUS?
A. ECLGSQ B.ECLGRQ C.ECLHSQ D.ECLQSQ
E.None of these

Q29. .If man is called girl, girl is called woman, woman is called boy, boy is called butler and butler is called rogue, who will serve in a restaurant ?
A. Butler B. Girl C. Man D. Woman E. Rogue

Q30. If IMPORT is written as USPQNJ, then how will CAPITAL be written in this code?
A.MBUJQBD B.KZSHOZB C.MUBJBDQ D.MBQJUBD
E.None of these

Q31. In a certain code 721 means' good college life' 526 means 'you are good' and 257 means 'life are good'. Which digits stand for 'you' in the code?
A.6 B.5 C.7 D.Can't be determined
E.None of these

Q32. If $\log x + \log y + \log z \geq 3$, then the smallest possible value of $x+y+z$ is
A. 10 B. 20 C. 30 D. 40 E. None of these

Q33. If $\log 2 = 0.3030$ and $\log 3 = 0.4771$, then find out the no of digits in 36^{42}.
A. 64 B. 65 C. 66 D. 67 E. None of these

Q34. Which one of the following is true
A. $\log_{23}499 = \log_{29}843$ B. $\log_{23}499 < \log_{29}843$
C. $\log_{23}499 > \log_{29}843$ D. Cannot be determined E. None of these

Q35. $\log_9 (3\log_2 (1 + \log_3 (1 + 2\log_2 x))) = 12$. Find x.
A. 4 B. 12 C. 1 D. 2 E. None of these

Q36. If $2^{2x+4} - 17 \times 2^{x+1} = -4$, then which of the following is true?
A. x is a positive value

B. x is a negative value
C. x can be either a positive value or a negative value
D. x is 0.
E. None of these

Q37. The value of $\left(\frac{1}{\log_3 60} + \frac{1}{\log_4 60} + \frac{1}{\log_5 60}\right)$ is:
A. 0 B. 1 C. 5 D. 60 E. None of these

Q38. Which of the following is true?
A. $7^{(3\wedge 2)} = (7^3)^2$ B. $7^{(3\wedge 2)} < (7^3)^2$ C. $7^{(3\wedge 2)} > (7^3)^2$ D. No Realtion
E. None of these

Q39. If $x = 4\sqrt{15}/(\sqrt{5} + \sqrt{3})$, then the value of $\frac{x+\sqrt{20}}{x-\sqrt{20}} + \frac{x+\sqrt{12}}{x-\sqrt{12}}$ is,
A. $\sqrt{3}$ B. 2 C. 1 D. $\sqrt{5}$ E. None of these

Q40. If M = (4^1/2) + (4^1/3) + (4^1/4) then the value of M is
A. less than 3 B. equal to 3 C. between 3 and 4 D. equal to 4
E. greater than 4

Q41. $\frac{2^{n+4} - 2(2^n)}{2.(2^{n+3})} + 2^{-3}$ is equal to
A. 2^{n+1} B. $-2^{n+1} + \frac{1}{8}$ C. $\frac{9}{8} - 2^n$ D. 1
E. None of these

Q42. Find the positive square root of $9 - \sqrt{56}$
A. $7-\sqrt{2}$ B. $\sqrt{2}-\sqrt{7}$ C. $\sqrt{7}-\sqrt{2}$ D. $7+\sqrt{2}$ E. None of these

Q43. The positions of how many digits in the number 2451479638 will remain same when the first half and the second half of the digits are arranged in ascending order separately?
A. Only one B. Two C. No Such Number D. Three
E. None of these

Q44. During a prize distribution ceremony, Vikram was ninth from the left while Janhvi was eighth from the right in the front row. If Hariom was thirteenth from the left and was exactly in the middle of Vikram and Janhvi in the same row then what was the total number of people in the front row?
A. 18 B. 19 C. 21 D. 24 E. None of these

Q45. In a north facing row of NCC Cadets, Trisha is 9th from the left end and Tina is 12th from the right end. There are 5 cadets between Trisha and Tanya who is equidstant to Tina. Find how many cadets are there in the row?
A. 34 B. 32 C. 31 D. 33 E. Can't be determined

Q46. In a class of students, Sairam's rank is 23rd from top in Maths, where as his rank is 27th from below in Physics. How many students are there in the class?
A. 45 B. 44 C. 60 D. Can't be determined
E. None of these

Q47. There are five books A, B, C, D and E. Book C lies above D, book E is below A, D is above A and B is below E. Which book is at the bottom?
A. E B. B C. A D. C E. None of these

Q48. Sonal is 22nd from the left and Rinku is 17th from the right. Counting from the left end, what will be the position of Pinku who is sitting exactly in the middle of Sonal and Rinku?
A. 27 B. 41 C. 37 D. Can not be determined
E. None of these

Q49. If A x B means A is to the south of B; A + B means A is to the north of B; A % B means A is to the east of B; A - B means A is to the west of B; then in P % Q + R - S, S is in which direction with respect to Q?
A. South-West B. South-East C. North-East D. North-West
E. None of these

Q50. A child went 90 m in the East to look for his father, then he turned right and went 20 m. After this he turned right and after going 30 m he reached to his uncle's house. His father was not there. From there he went 100 m to his north and met his father. How far did he meet his father from the starting point?
A. 80 m B. 100 m C. 140 m D. 260 m E. None of these

Q51. A horse is facing north. It turns 90 degrees in the clockwise direction, then 180 degrees in the anti-clockwise and then another 90 degrees in the same direction. Which direction is the horse facing now?
A. East B. South C. Southwest D. Southeast E. None of these

Q52. Early morning after sunrise, Karthik was standing infront of his house in such a way that his shadow was falling exactly behind him. He starts walking straight and walks 5 metres. He turns to his left and walks 3 metres and again turning to his left walks 2 metres. Now in which direction is he from his starting point?

A. West B. North-East C. East D. South-West E. None of these

Q53. In a class, 80% of students pass in maths exam and 70% of students pass in English exam. 10% of students fail in Both subjects and if 144 students pass in Both subejects. Find the total number of students.
A. 200 B. 240 C. 300 D. 480 E. None of these

Q54. A man had a certain amount with him. He spent 25% of that to buy an article and 10% of the remaining on clothes. Then he donated Rs 531.25. If he is left with Rs. 8000 the amount he spent on the clothes is
A. 816 B. 922.22 C. 947.916 D. 880.10 E. None of thes

Q55. Loss incurred by selling an article for Rs 800 is 20% more than profit gained by selling same article for Rs 1246 Find the cost price of this article(Approx)?
A. 1000 B. 1050 C. 1100 D. 1150 E. None of these

Q56. On selling 25 articles a person incurred loss equal to cost price of 5 articles. Find profit/loss.
A. 20% B. 24% C. 30% D. 48% E. None of these

Q57. A wooden block with dimensions 4x6x8 is cutted smaller cubes of equal size by making 6 cuts in total. How many smaller cubes will be formed?
A. 6 B. 192 C. 24 D. 48 E. None of these

Direction for-(Q58- Q59): A cube of 4 cm has been painted on its surfaces with different colours on each surface. Now the cube is cut into smaller cubes of side 1 cm each.

Q58. How many cubes will have no side painted?
A.18 B.16 C.22 D.8 E. None of these

Q59. How many cubes will have exactly two different coloured surfaces?

A.20 B.24 C.28 D.32 E. None of these

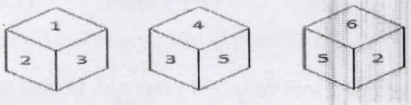
Direction for-(Q60-Q62): A cuboid shaped wooden block has 6 cm length, 4 cm breadth and 2 cm height. Two faces measuring 4 cm x 2 cm are coloured in black. Two faces measuring 6 cm x 2 cm are coloured in red. Two faces measuring 6 cm x 4 cm are coloured in green. The block is divided into smaller cubes of 1 cm.

Q60. How many cubes will have two different coloured surfaces?
A. 28 B. 16 C. 24 D. 32 E. None of these

Q61. How many small cubes will have one coloured surface ?
A. 16 B. 12 C. 6 D. 24 E. None of these

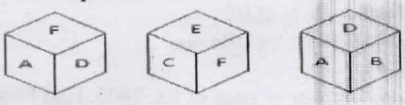
Q62. How many cubes will remain if the cubes having black coloured are removed ?
A. 24 B. 18 C. 12 D. 32 E. None of these

Q63. A dice with six faces is marked with six numbers 1, 2, 3, 4, 5 and 6 respectively. This dice is rolled three times and three positions are shown as:



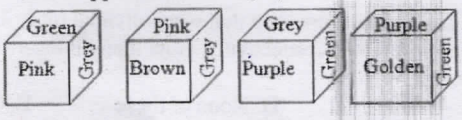
Find the number opposite to 1.
A. 2 B. 6 C. 5 D. 4 E. None of these

Q64. You are given three positions of dice then which face is opposite to the face with alphabet B?



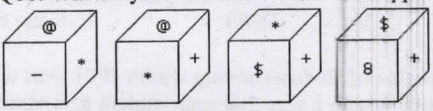
A. E B. F C. D D. A E. None of these

Q65. Refer to the following four positions of the dice and find out the color which is opposite the face grey?



A. Golden B. Purple C. Brown D. Green E. None of these

Q66. Which symbol will be on the face opposite to the face with symbol * ?



A. @ B. $ C. 8 D. + E. None of these

Q67. Namita has 4.2 kg of flour. She has been asked to make 5 cup cakes out of every ½ kg of flour. How many cup cakes can she bake out of the flour she has?
A. 21 B. 24 C. 30 D. 42 E. None of these

Q68. What is John's present age, if after 10 years his age will be 5 times his age 5 years back.
A. 6.2 years B. 7.7 years C. 8.7 years D. 10 years
E. None of these

Q69. Father is 3 times more aged than his daughter. If after 5 years, he would be 3 times of daughter's age, then further after 5 years, how many times he would be of his daughter's age?
A. 1 (½) times B. 2 times C. 2.5 times D. 3 times
E. None of these

Q70. A statue is 80 years old, and temple is 590 years old. In how many years will the temple be four times the age of the statue?
A. 60 years B. 70 years C. 80 years D. 90 years E. None of these

Q71. Rs. 600 are divided among A, B and C such that Rs. 40 more than 2/5 of A's share, Rs 20 more than 2/7 of B's share and Rs. 10 more than 9/17 of C's share all are equal. What is the share of B?
A. Rs. 280 B. Rs. 150 C. Rs. 240 D. Rs. 320

Q72. . I purchased a certain number of pens of Rs. 2800. Had I purchased 60 pens more, the cost of each pen would be Rs. 6 less. How many pens did I purchase?
A. 200 Pens B. 140 Pens C. 120 Pens D. None of these

Q73. A is the set of positive integers such that when divided by 2, 3, 4, 5, 6 leaves the remainders 1, 2, 3, 4, 5 respectively. How many integers between 0 and 100 belong to set A?
A. 1 B. 0 C. 2 D. None of these E. CND

Q74. The HCF of two numbers is 8. Which of the following can never be their LCM?
A. 24 B. 48 C. 56 D. 60 E. None of these

Q75. The maximum number of students among whom 1001 pens and 910 pencils can be distributed in such a way that each student gets same number of pens and same number of pencils, is
A. 91 B. 910 C. 1001 D. 1911
E. None of these

Q76. A number which when divided by 10 leaves a remainder of 9, when divided by 9 leaves a remainder of 8, and when divided by 8 leaves a remainder of 7 is

A. 1539 B. 539 C. 359 D. 1359

E. None of these

Q77. A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and c in 198 seconds, all starting at the same point. After what time will they again at the starting point ?

A.26min18secs B.45mins C.46 mins and 12 secs

D.42 mins 36 secs E. None of these

Q78. How many pairs of positive integers x, y exist such that HCF (x, y) + LCM (x, y) = 91?

A.2 B. 4 C. 6 D.8 E.10

Q79. How many even integers n , where $100 \leq n \leq 200$, are divisible by 7 or 9 ?

A. 37 B. 38 C. 39 D. 40 E. None of these

Q80. A number N= A127326 is divided by 66. What can be the difference of maximum and minimum value of A, where A is a two digit number?

A. 66 B. 89 C. 77 D. 12 E. None of these

Printed Pages : **University Roll No: .....**

End Term Examination, Even Semester 2018-19
B.Tech (CSE/CCV/DA), 2nd year, 4th Semester
BCSC0006 : Data Structure & Algorithms

Time: 3 Hour **Maximum Marks: 50**

Note: Write the complete code in one place neatly. Commenting the code is optional.

Section-A **7 x 5 = 35 Marks**

Q1. Define Data Structures. What is the difference between Linear and Non-linear data structures? What are the operations that can be performed on data structures.

Q2. Discuss the specification of a Binary Search Tree. Explain the different cases to delete an element from binary search tree.

Q3. What do you understand by stack. Explain the condition of Stack overflow and underflow.

Q4. Modify the body of the following method such that, if the input to the method is 33, 44, 55, 66 then, the method will return 4.

```
int size(Node list) {
// write your code here
}
```

Q5. Write program to implement doubly linked list provide following operations.

I. Insert at beginning
II. Insert at location
III. Remove from beginning
IV. Remove from location

Q6. Give ADT for Queue Data Structure. Discuss in brief any two applications of queue data structure.

Q7. Write code to Construct Binary search Tree from the following elements:-
12,8,25,14,9,6,18. Also show preorder, inorder and postorder traversal for the same.

Section-B

3 x 2 = 6 Marks

Q1. Sort the following data using merge sort and selection sort.
142,317,45,222,187

Q2. Define Sorting. What is the importance of Sorting?

Q3. Explain the term recursion with an example.

3 x 3 = 9 Marks

Q1. Explain with suitable example, BFS and DFS traversal of a graph.

Q2. What is AVL tree? Explain with suitable example the RR rotation and balance Factor.

Q3. Define the following Terms with respect to Trees:

A. Root
B. Subtree
C. Level of Node
D. Depth of Tree
E. Sibling
F. Forest

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End Term, Even Semester, 2018-19

B. Tech. (CSE/CCV/DA), II-Year, IV-Semester

BCSC 0010 : Discrete Mathematics

Time: 3 Hours

Max Marks: 50

Section-A

1. Attempt any two: **35 Marks**

a. Let $A=\{1,2,3,4,5\}\times\{1,2,3,4,5\}$. Let us define R on A by $(x1,y1)R(x2,y2)$ if $x1+y1=x2+y2$. Prove that the given relation is an equivalence relation.

b. Each student in a class of 40 plays at least one indoor game chess, carrom and scrabble. 18 play chess, 20 play scrabble and 27 play carrom. 7 play chess and scrabble, 12 play scrabble and carrom and 4 play chess, carrom and scrabble. Find the number of students who play chess and carrom but not scrabble.

c. Let $A = \{0, 1, 2, 3, 4\}$ and $A - \{0\} = \{1, 2, 3, 4\}$. Can you define a function $f : A - \{0\} \rightarrow Z$ as follows: For every $x \in A - \{0\}$, $f(x)$ is an integer y such that $(xy) \bmod 5 = 1$. Construct an example of such f.

2. Attempt all:

a. In how many ways can we write 20 as a sum of 3 non-negative integers?

b. In how many ways can the letters of the word 'MILLIMICRON' be arranged?

c. How many integers from 100 to 999 must be picked in order to be sure that atleast 2 of them have a digit in common?

3. Attempt all:

a. A sequence of numbers $u_1,u_2,u_3,...$ is defined as $u_{n+1}=4u_n+2$, where $u_1=2$. Find the solution of this recurrence relation.

b. Find the closed form of generating function for the following sequence:

1,0,-1,0,1,0,-1,0,1,0,-1,0,.....

4. Attempt all:

a. Prove that $\sqrt{5}$ is an irrational number.

b. If from a lottery of 30 tickets, marked 1 to 30, four tickets are to be drawn. What is the probability that those marked 1 and 2 are among them?

5. Attempt any one:

a. Show that the premises "If I study then I will not fail in Mathematics". "If I do not hang out with friends, then I will study". "But I failed in mathematics" imply the conclusion, "I must have hung out with friends." Using inference theory/rules on propositions.

b. Show that the premises "Everyone in this class is happy" and "James is a student in this class" imply the conclusion

"James is happy." Using predicates and inference rules.

6. Show that the set of integers with composition o and *, represented as (Z, o, *) is a ring where the composition o and * is defined by

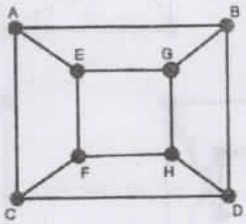
   $a \circ b = a + b + 1$ and $a * b = ab + a + b$

7. Attempt any two:
   a. Which of the graphs K4 and Q3 is/are planar? Give reasons for your answer.
   b. Let G be a graph with n vertices and G' be the complement of G, then find no of edges in E(G) + E(G').
   c. State and prove handshaking theorem on graphs.

Section-B

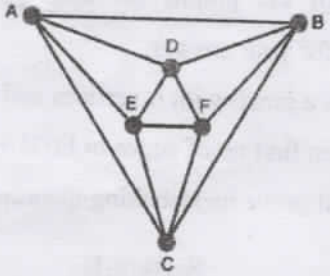
(a) Attempt all questions: **(3x2=6)**

1. State and prove Lagrange's theorem on subgroups.
2. Find order of each element 1, 2, 3, and 4 in the group $(G,+_5)$ where $G = \{0,1,2,3,4\}$ and $a +_5 b = (a+b) \bmod 5$.
3. Find out whether the graph given below is bipartite or not? If yes, Give two distinct set of vertices.

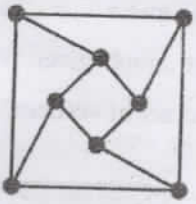


(b) Attempt all questions: **(3x3=9)**

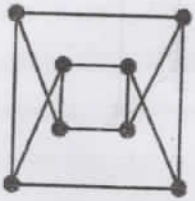
1. If the binary operation * is defined as a*b = a+b-2, prove that operation * has an identity. Also find the inverse of an element a in R with respect to this binary operation *.
2. Given the graph below, find
   a. Euler path and circuit
   b. Hamiltonian path and circuit



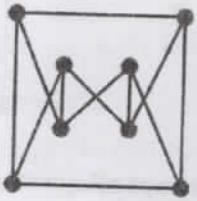
3. Consider the following three graphs. Find out which of these are isomorphic to each other? Also, give the corresponding mapping of vertices.



G



G_1



G_2

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End Term Examination, Even Semester 2018-19
B. Tech(CS), II year, IV Semester
BCSC 0008- Computer Network

Time: 3 Hour Maximum Marks: 50

Section-A

Attempt all question. **7 x 5 = 35 Marks**

(i) What is CIDR? Classless Inter-domain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries: **Prefix Output Interface Identifier**

Prefix	**Output Interface**	**Identifier**
131.16.0.0/12		3
131.28.0.0/14		5
131.19.0.0/16		2
131.22.0.0/15		1

On which interface this packet will be forwarded. Justify your answer with suitable reason.

(ii) Compare and contrast byte-stuffing and bit-stuffing with example. Which technique is used in byte-oriented protocols? Which technique is used in bit-oriented protocols?

(iii) Compare Go-Back-N protocol with Selective-Repeat protocol in terms of efficiency, number of retransmissions and window size (both sender and receiver). In GB3 if every 5th packet is lost and we need to send 10 packets so how many retransmissions are required?

(iv) How pure ALOHA network differs from slotted ALOHA? What is the vulnerable time of each? We have a pure ALOHA network with 100

stations. If Tfr = 1 μ s, what is the number of frames/s each station can send to achieve the maximum efficiency.

(v) Write the responsibilities of network and transport layer in OSI reference model and compare OSI model with TCP/IP model.

(vi) Discuss the role of router and gateway in the network. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.

(vii) What do you understand by the term cryptography? How symmetric key cryptography is differing from asymmetric key cryptography? Discuss the substitution cipher and transposition cipher.

Section-B

(a) Attempt all question. **3 x 2 = 6 Marks**

(i) How token bucket algorithm is differing from leaky bucket?

(ii) Generate the public key and secrete key for the following prime numbers using RSA algorithm, P=3, Q=11take E=5.

(iii) What is domain name? How is a domain name translated into equivalent IP address?

(b) Attempt all question **3 x 3 = 9 Marks**

(i) Consider an instance of TCP's Additive Increase Multiplicative Decrease(AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

(ii) IPv4 has some deficiencies that make it unsuitable for the fast-growing Internet and provide space for IPv6; comment on it, also explain the frame format of IPv6 datagram.

(iii)Explain the connection establishment and termination process in TCP.

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END -Term Examination, Even Semester 2018-19
B.TECH, (CS/DA/CCV), II YEAR, IV SEMESTER
BCSC 0007: Introduction to Microprocessor

Time: 3 Hour **Maximum Marks: 50**

Section – A

Attempt all questions **7 X 5 = 35 Marks**

Q1. Write a program in 8085 to find out the sum of numbers in a given series. The length of the series is in memory location 4200H and the series begins from memory location 4201H. Consider the sum to be 8 bit number. So, ignore carries. Store the sum at memory location 4300H.

Q2. Draw the pin diagram of 8086 microprocessor and explain the name of signal associated with maximum mode operation of 8086 microprocessor.

Q3. Sketch and explain the programming model of 8085.Also explain the need of demultiplexing of bus AD7-AD0 in 8085.

Q4. Explain mnemonic code with example

a. RAR b. DAA c. PUSH d. SHLD e. CPI Data

Q5. Highlight the use of addressing modes used in the following instruction.

a. ADI 23H **b.** MOV M, A **c.** LDA D500H
d. LXI D, 8C50H **e.** CMA

Q6. Draw the internal architecture of 8255. List the operating modes of 8255. Specify the bit of a control word for 8255, which differentiate between I/O mode and BSR mode?

Q7. Explain the purpose of peripheral interfacing in detail. What are the differences between interrupt driven I/O and DMA I/O?

OR

What is the necessity of the programmable interval timer? Discuss the organization of Timer with a functional block diagram.

Section – B

(a) **Attempt all questions** **3 X 2 =6 Marks**

Q1. How many 8259s IC can be connected in cascaded mode, show their Cascading structure?

Q2. Find the data direction and modes of operation of the 8255 ports, if the control word written into it is OAH.

Q3. Discuss the Command and Mode register configuration of 8237 DMA controller.

(b) **Attempt all questions** **3 X 3 =9 Marks**

Q1. Explain the data transfer types supported by 8237 and also explain the current and base register in 8237.

Q2. With the help of timing diagram explain mode0 and mode1 operation of 8253 timer/counter.

Q3. Use schematic diagram to discuss the internal architecture of PIC 8259.

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End Term Examination, Even semester 2018-19

B.Tech. II Year/ IV Semester

Software Engineering (BCSC 0009)

Uni. Roll No: .....

Time: 3 Hrs. Total Marks: 50

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions (7x5=35)

I. Consider a mail order company that distributes CDs and tapes at discount prices to its members. When an order processing clerk receives an order form, she verifies that the sender is a member by checking the MEMBER FILE.
If the sender is not a member, the clerk returns the order along with a membership application form. If the customer is a member, the clerk verifies the order item data by checking the ITEM FILE.
Then the clerk enters the order data and saves it to the DAILY ORDERS FILE. At the same time the clerk also prints an invoice and shipping list for each order, which are forwarded to the ORDER FULFILLMENT DEPARTMENT for processing there.
Consider the above scenario and create DFD level-0, level-1, level-2.

II. Consider a C project with the following parameters.
(i) External Inputs:
(a) 10 with low complexity
(b) 15 with average complexity
(c) 17 with high complexity
(ii) External Outputs:
(a) 6 with low complexity
(b) 13 with high complexity
(iii) External Inquiries:
(a) 3 with low complexity
(b) 4 with average complexity
(c) 2 high complexity

(iv) Internal logical files:
(a) 2 with average complexity
(b) 1 with high complexity
(v) External Interface files:
(a) 9 with low complexity
In addition to above, system requires
i. Significant data communication
ii. Performance is very critical
iii. Designed code may be moderately reusable
iv. System is not designed for multiple installation in different organizations.
Other complexity adjustment factors are treated as average. Calculate the Average staff size and Productivity of the project using COCOMO Model.

III. Define the role of structure Design methodology in designing a software. Explain all four steps of SDM for given scenario:
Company have to design a software system to determine the number of unique words in the file given by user.

IV. Compute vocabulary, program length, volume of the given code:

```
void arrange_data (int a, int n)
{
int i, j, t;
    if( n < 2 )
    return;
 for ( i=0 ; i <n-1; i++ )
  {
   for ( j=i+1 ; j < n ; j++ )
     {
       if( a[i] > a[j] )
         {
          t= a[i]; a[i] = a[j]; a[j] = t;
         }
     }
  }
}
```

V. Consider the fallowing program and Draw flow graph, DD path graph, identify independent paths, Write the test cases, Calculate cyclomatic complexity by all four methods.

```
#include <stdio.h>
Void main ()
{
  int a = 100;
  if( a == 10 )
{
     printf("Value of a is 10\n" );
  }
else if( a == 20 )
 {
    printf("Value of a is 20\n" );
  }
 else if( a == 30 )
 {
    printf("Value of a is 30\n" );
  }
 Else
 {
     printf("None of the values is matching\n" );
  }
  printf("Exact value of a is: %d\n", a );
  getch();
}
```

VI. Explain Equivalence Class Testing Method.
Consider the following scenarios and write the test cases according to Equivalence Class Testing Method for each scenarios.
Assume, we have to test a field which accepts Age 18 – 56 for man and Age 20-52 for women.
Assume, we have to test a filed which accepts a Mobile Number of ten digits.
Suppose you have very important tool at office, accepts valid User Name and Password field to work on that tool, and accepts minimum 8 characters and maximum 12 characters. Valid range 8-12, Invalid range 7 or less than 7 and Invalid range 13 or more than 13.

VII. Define Categories of Software Maintenance. Which category of software maintenance consumes maximum cost of maintenance and why? Explain Taute Maintenance Model with the help of diagram.

Section B

Note: Attempt All Question. (3x2=6)

Part-1

I. Differentiate between Alpha and Beta testing

II. What is the role of Driver and stub in Unit Testing? Explain with diagram.

III. What is the need of Reverse Engineering? Explain with diagram.

Part-2 (3x3=9)

I. Explain Mutation Testing. Consider a project of size 500KLOC needed maintenance, under maintenance company is going to delete 30KLOC and add 70KLOC in project. Compute an estimate for annul maintenance effort.

II. Explain CMM model and enlist the KPA of level 3 and level 5.

III. Explain ISO-9126 software quality model with the help of diagram.

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End Term, Even Semester, 2018-19
B. Tech. (CS) II-Year, IV-Semester
BCHS 0201 : Environmental Studies

Time: 3 Hours Max Marks: 50

Section A

Attempt all questions [5x7=35]

Ques 1. [2.5 × 2]

(i) Why is environmental awareness mandatory for all the people in the society?

(ii) How does the earth manage its radiation balance so as to maintain an average surface temperature of 15°C?

Ques 2. [2.5 × 2]

(i) Describe functions of atmosphere, hydrosphere and lithosphere.

(ii) What is the future threat to the availability of fresh water on earth?

Ques 3. [2.5 × 2]

(i) Define ecological pyramid. Draw different types of ecological pyramids with suitable example.

(ii) What is fossil fuel and what are the harmful effects of the combustion of fossil fuel.

Ques 4. [2.5 × 2]

(i) Describe the impact of modern agricultural practice on the environment.

(ii) What are the major threats to the biodiversity of India?

Ques 5. [2.5 × 2]

(i) Explain the difference between BOD, COD and DO of the water bodies.

(ii) Write a note on common air pollutants and their effects on human health.

Ques 6. [2.5 × 2]

(i) Write a note on ecological foot print.

(ii) Discuss the salient features of the wild life protection act 1972.

Ques 7. [5]

What is meant by environmental impact assessment? Describe its process in detail.

Section B

Attempt all questions

Ques 1. [3x2 =6]

(i) How environmental ethics contribute to solve the environmental problems?

(ii) What are the major obstacles in the path of sustainable development?

(iii) Enumerate various efforts which are being made by NGOs towards environmental concerns.

Ques 2. [3x3 = 9]

(i) "Environmental legislation can be made effective only by making people aware of importance of environmental laws" justify the statement.

(ii) Discuss the power of central government under environmental protection act 1986.

(iii) Discuss the role of citizens in environment protection.